

Installation and User Guide

Deliverable 2 final guide - Smart Campus Event Planner, Team 16

1. System requirements

- Node.js 22 or newer, which includes npm.
- A modern browser such as Chrome, Edge, Firefox, or Safari.
- No separate database server is required. The application uses a local SQLite database.

2. Installation

Unzip the submission, open a terminal in the project folder, and run:

```
npm install
```

This installs Express, SQLite support, bcrypt, session support, and the development tools listed in package.json. Do not open the HTML files with Live Server because the application requires its Node server and API routes.

3. Database setup and startup

Start the application with:

```
npm start
```

The start command runs the database initializer before the server. On the first run it creates database/campus.db, creates the tables, and inserts demonstration users, events, and registrations. Re-running the command does not duplicate the seed data.

Open the application at:

```
http://127.0.0.1:3000
```

For development with automatic server restarts, use npm run dev. Stop the server with Control+C.

4. Demonstration accounts

Role	Email	Password	Recovery answer
Student	student@demo.com	student123	Montreal
Admin	admin@demo.com	admin1234	Concordia

5. Student user guide

Create an account

- Select Sign up from the login page.
- Enter a full name, valid email, password of at least eight characters, and matching confirmation.
- Choose a security question and enter an answer for password recovery.
- After registration, return to Login and use the new credentials.

Browse and filter events

- Open Events from the navigation bar.
- Search by event title, description, location, or organizer.
- Filter by category, exact date, or event status.
- Select View Details to see the complete event information, organizer, registration count, and remaining seats.

Register and cancel

- On an open future event, select Register. The server rejects duplicate, full, cancelled, disabled, completed, or past events.
- Open My Registrations to see all personal registration records.
- Use All, Upcoming, or Past to filter the list.
- Select Cancel Registration for an eligible future registration and confirm the action.

Dashboard and profile

- The dashboard shows registered, upcoming, attended, and cancelled totals from the database.
- Participation progress and category totals update from the student's records.
- Open Profile to update the logged-in student's name or email.

Recover a password

- Select Forgot password? on the login page.
- Enter the account email, choose its security question, and provide the answer.
- After successful verification, enter and confirm a new password.

6. Admin user guide

- Log in with the admin demonstration account to open the admin dashboard.
- Use Create Event and complete every required field. Dates cannot be in the past, capacity must be positive, and the end time must be after the start time.
- Use Manage Events to search, edit, cancel, reopen, disable, enable, or delete events.
- The admin dashboard lists registered students and supports marking each active registration as attended or absent.
- Dashboard statistics show event totals, active registrations, full-capacity events, attendance, capacity percentage, attendance rate, category totals, and the most popular category.

7. Troubleshooting

- If node or npm is not recognized, install a current Node.js release and reopen the terminal.
- If dependencies are missing, run npm install again from the folder containing package.json.
- If port 3000 is already in use, stop the other process before running npm start.
- If the database needs a fresh demonstration state, stop the server, remove database/campus.db, and run npm start again.
- Use npm test to run the automated authentication, registration, capacity, role, CRUD, attendance, and statistics checks.